

PAPER_1209-X: UQFF Climate & Atmosphere Unified Proof Set

—

Five Exact Closures Including CO₂, Salinity, Albedo, Greenhouse Effect, Ozone

Daniel T. Murphy

May 16, 2026

Locked Primitives

$F_{\text{TRZ}} = \frac{1}{10}$, $\Phi_{\text{res}} = \frac{5}{6}$, $[\text{SSq}] = \frac{57}{100}$, $K_{\text{Mex}} = \frac{25}{12}$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

Closures S553–S562

S553 — CO₂ atmospheric concentration (ppm), target 420 [EXACT]

$$A_5 D_{\text{phys}} + D_{\text{crit}} D_{\text{BSFG}} + D_{\text{BSFG}} D_{\text{phys}} = 240 + 156 + 24 = 420.$$

S554 — Adiabatic lapse rate (K/km), target 6.5 [0.21%]

$$D_{\text{BSFG}} + [\text{SSq}] - F_{\text{TRZ}} \Phi_{\text{res}} = 6.4867.$$

S555 — Atmospheric scale height (km), target 8.5 [0.71%]

$$2D_{\text{phys}} + [\text{SSq}] - F_{\text{TRZ}}^2 = 8.56.$$

S556 — Solar constant (W/m²), target 1361 [0.0022%]

$$A_5^2 F_{\text{TRZ}} D_{\text{phys}} - N_{\text{ch}} \text{SO5} + \text{SO5} + [\text{SSq}] + F_{\text{TRZ}} D_{\text{phys}} = 1360.97.$$

S557 — Ocean salinity (ppt), target 35 [EXACT]

$$D_{\text{crit}} + N_{\text{ch}} = 35.$$

S558 — Standard atmospheric pressure (kPa), target 101.325 [0.0049%]

$$\text{SO5}^2 + [\text{SSq}] + \Phi_{\text{res}} - F_{\text{TRZ}} \Phi_{\text{res}} = 101.32.$$

S559 — Earth Bond albedo, target 0.30 [EXACT]

$$3F_{\text{TRZ}} = 0.30.$$

S560 — Greenhouse effect (K), target 33 [EXACT]

$$D_{\text{crit}} + N_{\text{ch}} - K_{\text{Mex}} + F_{\text{TRZ}}\Phi_{\text{res}} = 35 - \frac{25}{12} + \frac{1}{12} = 33.$$

S561 — Tropopause height (km), target 11 [0.38%]

$$\text{SO5} + \Phi_{\text{res}} + K_{\text{Mex}}F_{\text{TRZ}} = 11.042.$$

S562 — Total ozone column (DU), target 300 [EXACT]

$$A_5D_{\text{phys}} + \text{SO5}D_{\text{BSFG}} = 240 + 60 = 300.$$

Summary

ID	Quantity	Target	Error
S553	CO ₂ ppm	420	EXACT
S554	Lapse rate K/km	6.5	0.21%
S555	Scale height km	8.5	0.71%
S556	Solar const W/m ²	1361	0.0022%
S557	Salinity ppt	35	EXACT
S558	Pressure kPa	101.325	0.0049%
S559	Bond albedo	0.30	EXACT
S560	Greenhouse K	33	EXACT
S561	Tropopause km	11	0.38%
S562	Ozone DU	300	EXACT

Highlights. Tier X delivers **five exact closures** (S553, S557, S559, S560, S562) — the most exact tier in the program. The greenhouse effect $33 = D_{\text{crit}} + N_{\text{ch}} - K_{\text{Mex}} + F_{\text{TRZ}}\Phi_{\text{res}}$ exhibits perfect arithmetic ($35 - 24/12 = 33$). Bond albedo as $3F_{\text{TRZ}}$ and salinity as $D_{\text{crit}} + N_{\text{ch}}$ are the cleanest single-term closures yet recorded. Solar constant matches to 22 ppm, atmospheric pressure to 49 ppm. Cumulative: 267 closures; 51 lockings (5 new from exact climate closures).